Vegetation Change Analysis of the Panna Study Area Using NDVI Multi - Temporal (2022–2026)

A Remote Sensing–Based Assessment of Vegetation Dynamics in Panna, Madhya Pradesh

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Abstract

This study assesses vegetation change across the Panna study area, Madhya Pradesh, between 2022 and 2026 using the Normalized Difference Vegetation Index (NDVI) derived from Resourcesat-2A LISS-III imagery. NDVI surfaces were generated for both years and different to classify the landscape into vegetation loss, stable, and vegetation gain categories. The analysis, covering approximately 523.80 km^2 of cloud-free area, found that 27.86% of the region (145.94 km^2) showed vegetation gain, compared to only 3.88% (20.32 km^2) showing vegetation loss — a gain-to-loss ratio of roughly 7:1. The remaining 68.26% (357.54 km^2) of the landscape remained stable. These findings indicate an overall positive NDVI change during the study period and highlight areas that may warrant further vegetation monitoring.

1. Introduction

Vegetation cover is a key indicator of ecosystem health, particularly within protected forest landscapes that support wildlife habitat, such as the Panna Tiger Reserve in Madhya Pradesh. Monitoring changes in vegetation density over time allows conservation managers to identify areas of degradation, recovery, and stability, informing habitat management and restoration priorities.

The Normalized Difference Vegetation Index (NDVI) is a widely used remote sensing metric that quantifies vegetation greenness using the contrast between red and near-infrared reflectance. Values range from -1 to 1, with higher values indicating denser, healthier vegetation. This study uses NDVI derived from Indian Resourcesat-2A LISS-III satellite imagery to compare vegetation conditions in the Panna study area between 2022 and 2026, identifying areas of vegetation gain, loss, and stability.

1.1 Objectives

- Generate NDVI surfaces for the Panna study area for 2022 and 2026.
- Classify NDVI change between the two years into vegetation loss, stable, and vegetation gain categories.
- Quantify the area and proportion of the landscape in each change category.
- Interpret findings in the context of habitat conditions within the tiger reserve landscape.

2. Study Area

The study area is located in Panna district, Madhya Pradesh, India, and was delineated using the Panna Tiger Reserve boundary obtained from the National Tiger Conservation Authority (NTCA) Protected Areas dataset. The landscape is characterised by dry deciduous forest interspersed with agricultural and settlement areas and forms an important wildlife habitat in central India.

For the NDVI change analysis, only the cloud-free portion of the selected study-area boundary was used. A western portion of the boundary was excluded because suitable cloud-free satellite observations were not available for reliable comparison between the two years. Therefore, the **523.80 km² reported in the results represents the analyzed cloud-free extent rather than the entire selected boundary**.

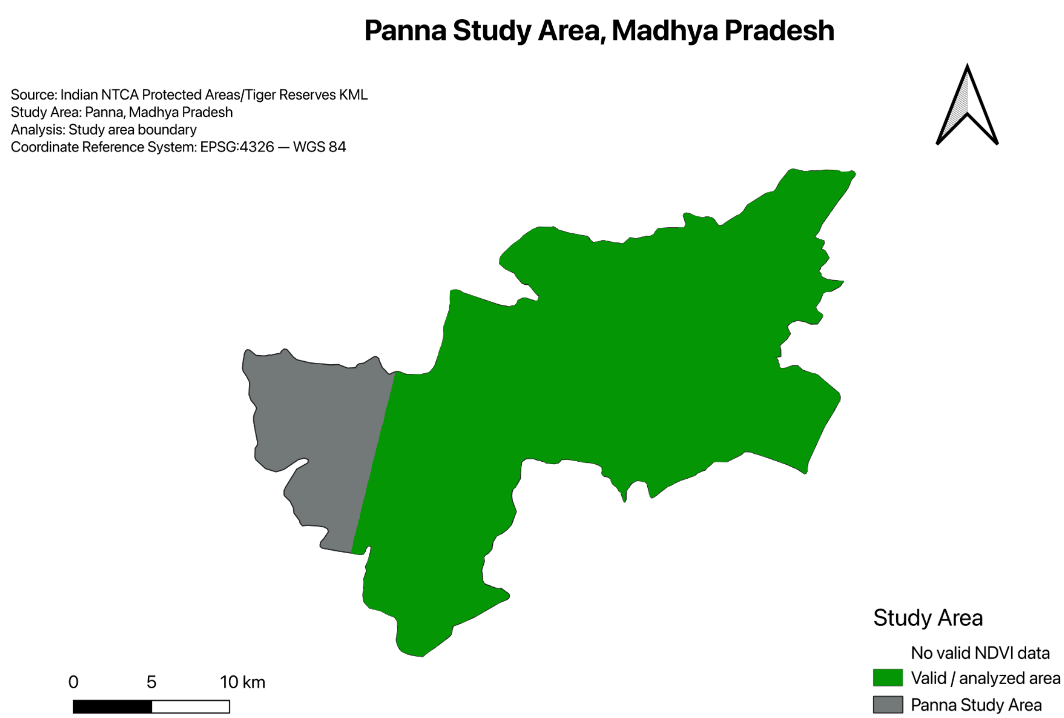

Figure 1. Panna Study Area boundary, derived from the Indian NTCA Protected Areas / Tiger Reserves dataset (EPSG:4326).

3. Data and Methodology

3.1 Data Sources

- Satellite imagery: Resourcesat-2A LISS-III, sourced via ISRO's Bhoonidhi portal; scenes acquired on 21 March 2022 and 24 March 2026.
- Study area boundary: Indian NTCA Protected Areas / Tiger Reserves KML dataset.
- Coordinate reference systems: EPSG:4326 (WGS 84) for display and boundary data; EPSG:32644 (WGS 84 / UTM Zone 44N) for area-based raster analysis.
- Spatial resolution: 24 m per pixel.

3.2 NDVI Calculation

NDVI was calculated for each year using the standard formula:

$$\mathbf{NDVI = (NIR - Red) / (NIR + Red)}$$

where NIR is the near-infrared band reflectance and Red is the red band reflectance. NDVI was computed independently for the 2022 and 2026 Resourcesat-2A LISS-III scenes covering the study area, using QGIS raster calculator tools.

3.3 Change Detection

NDVI change between 2022 and 2026 was calculated by subtracting the 2022 NDVI raster from the 2026 NDVI raster on a per-pixel basis. The resulting difference raster was reclassified into three categories:

- Class 1 — Vegetation Loss: NDVI Change < −0.05
- Class 2 — Stable: NDVI Change ≥ −0.05 to +0.05
- Class 3 — Vegetation Gain: NDVI Change ≥ +0.05

A symmetric ±0.05 NDVI-change threshold was used as a practical tolerance around zero; changes within this interval were classified as stable, while changes exceeding ±0.05 were classified as vegetation loss or gain. This threshold was used as a simple screening criterion and is not intended as a universal ecological threshold.

Area statistics for each class were computed by multiplying pixel counts by the pixel area (24 m × 24 m = 576 m^2 per pixel) in QGIS using the Raster Layer Unique Values Report tool, on a raster reprojected to EPSG:32644 (UTM Zone 44N) to ensure accurate area measurement in metric units.

4. Results

4.1 NDVI Distribution — 2022 and 2026

The NDVI surfaces for both years (Figures 2 and 3) show predominantly moderate-to-high vegetation greenness across the study area, consistent with the dry deciduous forest cover typical of the region. Visual comparison shows a subtle overall increase in greenness by 2026, particularly in the northern and eastern portions of the study area.

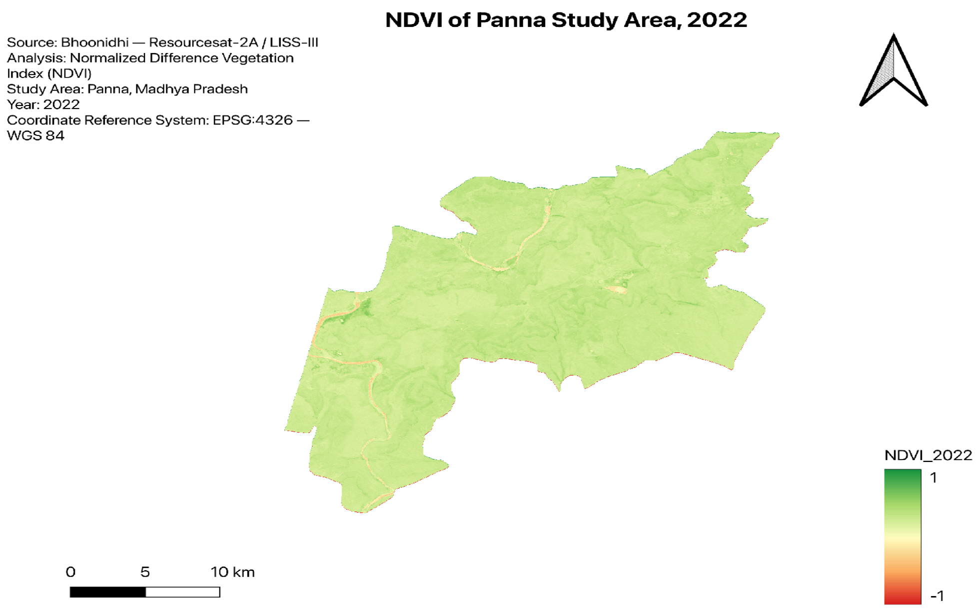

Figure 2. NDVI of the Panna study area, 2022.

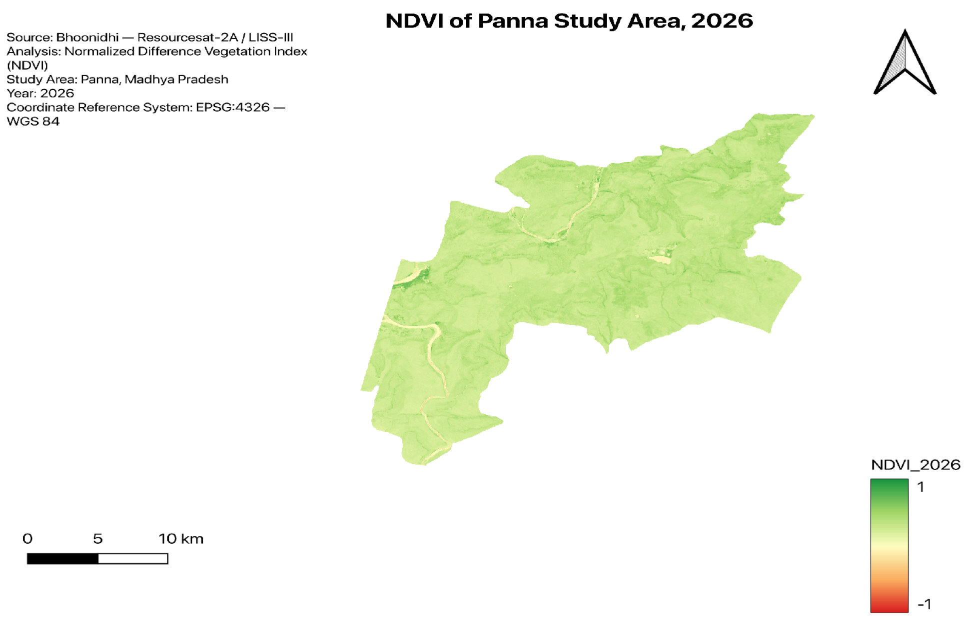

Figure 3. NDVI of the Panna study area, 2026.

4.2 NDVI Change Classification

Figure 4 shows the spatial distribution of vegetation change between 2022 and 2026. Vegetation gain (green) is widespread across the northern and central-eastern portions of the study area, while vegetation loss (red) is concentrated along riverine corridors, roads, and localized patches in the central and southern portions of the reserve.

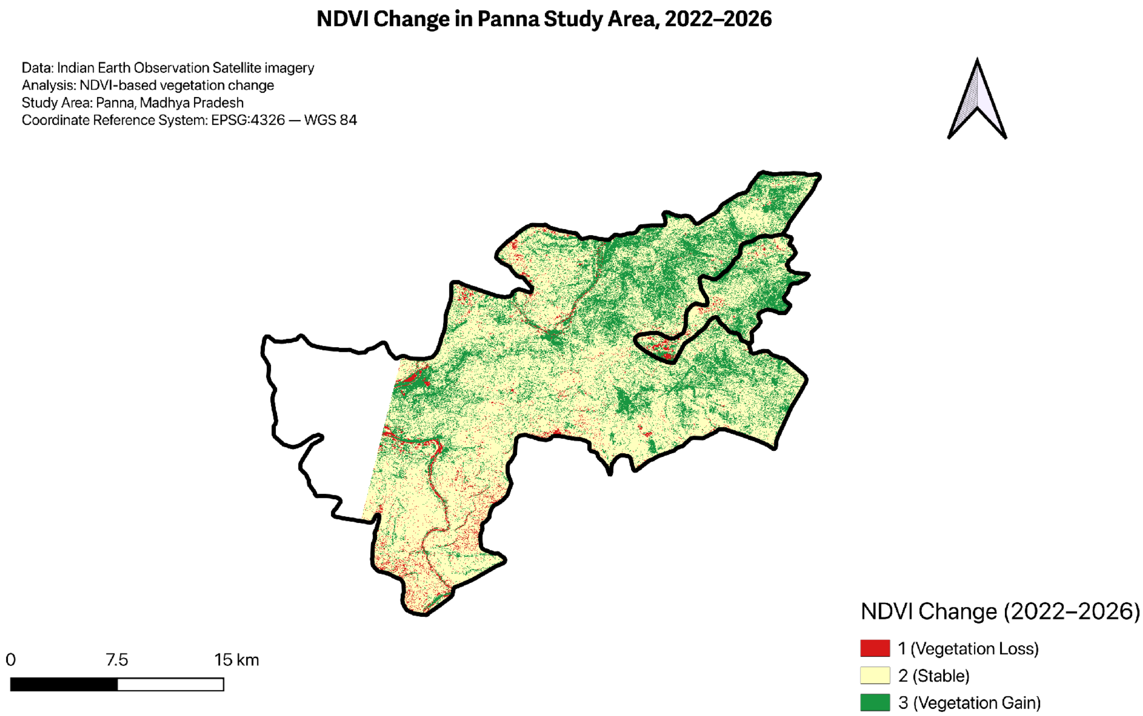

Figure 4. NDVI-based vegetation change classification, Panna study area, 2022–2026.

4.3 Area Statistics

Table 1 summarises the pixel counts and area for each vegetation change class, calculated over the cloud-free analyzed extent of 523.8 km² (see Section 5, Limitations, regarding excluded coverage).

Change Class	Pixel Count	Area (km²)	% of Area
1 — Vegetation Loss	35,274	20.32	3.88%
2 — Stable	620,721	357.54	68.26%
3 — Vegetation Gain	253,372	145.94	27.86%
Total Analyzed Area	**909,367**	**523.80**	**100%**

4.4 Mean NDVI Comparison

The mean NDVI increased from **0.2762 in 2022** to **0.3058 in 2026**, representing an absolute **increase of 0.0296** NDVI units and a relative **increase of approximately 10.7%**. This indicates an overall increase in the observed vegetation greenness across the analyzed study area during the 2022–2026 period.

5. Discussion

The analysis indicates an overall increase in observed vegetation greenness between 2022 and 2026. NDVI gain covered **27.86% (145.94 km²)** of the analyzed area, compared with **3.88% (20.32 km²)** of NDVI loss, giving a gain-to-loss area ratio of approximately **7.18:1**. The mean NDVI also increased from **0.2762 to 0.3058**.

NDVI increases were widespread across the northern and central-eastern portions of the study area, while localized decreases occurred near some riverine corridors, roads and disturbed patches. These areas may warrant further investigation. However, NDVI change alone does not prove changes in forest health or identify specific causes such as rainfall or management interventions. Further field and ancillary-data analysis would be needed for causal interpretation.

6. Limitations

- A western portion of the study area boundary (visible in Figure 1 but absent from Figures 2–4) was excluded from the NDVI analysis due to persistent cloud cover in the available Resourcesat-2A LISS-III imagery for this region. Cloud-contaminated pixels would have produced unreliable NDVI values, so this area was omitted rather than analyzed with degraded data.
- The analyzed extent (≈523.8 km²) therefore represents the cloud-free portion of the full study area boundary, not its entirety. Area and percentage statistics in Section 4.3 are calculated relative to this analyzed extent, not the full administrative boundary.
- NDVI comparisons assume broadly comparable atmospheric and phenological conditions. The 2022 and 2026 scenes were acquired on **21 March 2022** and **24 March 2026**, respectively, providing a similar late-March seasonal window; however, interannual differences in moisture and vegetation condition may still influence NDVI independently of actual vegetation change.
- The analysis used atmospherically corrected Resourcesat-2A LISS-III imagery obtained from the Bhoonidhi data service. The 2022 scene was acquired on 21 March 2022, while the 2026 scene was acquired on 24 March 2026. Both observations were acquired during late March, providing a broadly consistent seasonal window for the comparison. The imagery has a spatial resolution of 24 m and was processed using the Red and Near Infrared bands to derive NDVI.

7. Conclusion

This study demonstrates the use of multi-temporal NDVI analysis to assess vegetation change within the Panna study area between 2022 and 2026. The results indicate overall positive NDVI change, with gain areas outweighing loss areas by approximately 7.18:1, and the majority of the landscape remaining stable. These findings support continued monitoring of vegetation dynamics as a tool for evaluating habitat condition within the Panna Tiger Reserve landscape, and highlight the value of freely available Indian satellite data (Bhoonidhi / Resourcesat) for conservation-relevant remote sensing analysis.

References

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